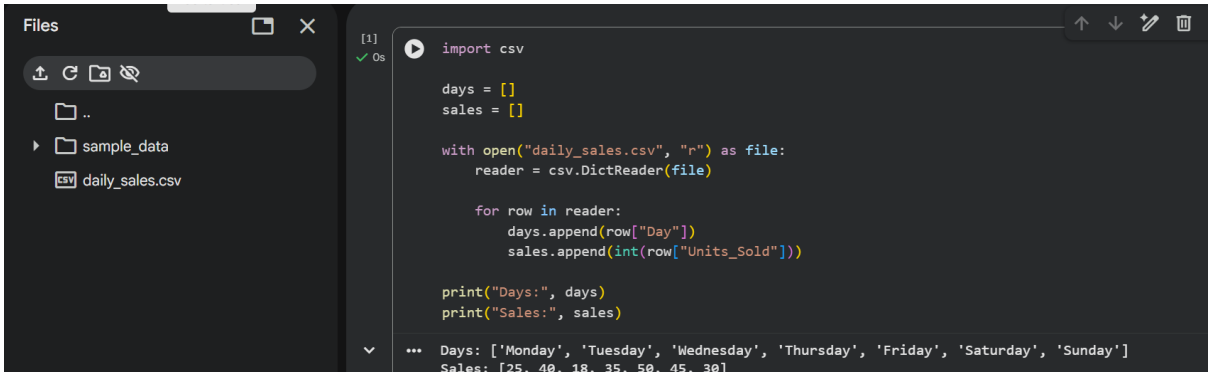
Without NumPy: Use Python's built-in csv module, lists, loops, and basic arithmetic.

With NumPy: Use NumPy arrays and built-in numerical functions.

NumPy — Numerical computing

Used for numerical arrays, mathematical calculations, statistical operations, and efficient calculations across large datasets.

Without Numpy:




```
total = sum(sales)
minimum = min(sales)
maximum = max(sales)
average = total / len(sales)

print("Total units sold:", total)
print("Minimum daily sales:", minimum)
print("Maximum daily sales:", maximum)
print("Average daily sales:", average)
```

```
Total units sold: 243
Minimum daily sales: 18
Maximum daily sales: 50
Average daily sales: 34.714285714285715
```

With Numpy:

Create a NumPy array

```
daily_units = np.array([5, 7, 4, 8, 6])

print("Daily laptop units:", daily_units)
print("Data type:", daily_units.dtype)
print("Number of values:", daily_units.size)
```

```
Daily laptop units: [5 7 4 8 6]
Data type: int64
Number of values: 5
```

Calculate sales statistics with NumPy

```
print("Total units sold:", np.sum(daily_units))
print("Minimum daily sales:", np.min(daily_units))
print("Maximum daily sales:", np.max(daily_units))
print("Average daily sales:", np.mean(daily_units))
print("Median daily sales:", np.median(daily_units))
print("Standard deviation:", round(np.std(daily_units), 2))
```

```
Total units sold: 30
Minimum daily sales: 4
Maximum daily sales: 8
Average daily sales: 6.0
Median daily sales: 6.0
Standard deviation: 1.41
```

Calculate revenue with NumPy

```
laptop_price = 40000

daily_revenue = daily_units * laptop_price

print("Daily revenue:", daily_revenue)
print("Total revenue:", np.sum(daily_revenue))
print("Average daily revenue:", np.mean(daily_revenue))
```

```
Daily revenue: [200000 280000 160000 320000 240000]
Total revenue: 1200000
Average daily revenue: 240000.0
```